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# Laser Descaling Robot for Long Range Scale Removal Applications

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## Abstract

Scale deposits gradually accumulate within flowlines, often resulting in obstructions. At present, the descaling process primarily relies on chemical cleaning, pigging, or hydroblasting methods. However, these approaches are range limited, require complex step-up, and carry the risk of potential damage to the inner pipe surface. To overcome these issues, we developed a laser descaling robot that can move within the flowline while carrying a high-power laser that can remove materials in the scale buildup.

The designed robot for descaling operations utilizes an in-pipe crawler designed to be centralized in the pipe. The laser collimator is affixed to a mount and positioned near the edge of the pipe. The robot's mount is attached to the end of the crawler and is equipped with a motor that facilitates the rotation of the collimator. Additionally, the robot is equipped with various hoses and accessories necessary for laser descaling activities. Furthermore, for real-time monitoring of the descaling process, a camera is affixed to the laser mount, ensuring that the operation can be closely observed as it unfolds.

A test setup is constructed extending over 60 m with varying pipes' internal diameters between 7.65- and 8-inch. Additionally, the pipe setup features a 10 m radius curvature to verify the descaling robot's capability to navigate curved paths. Furthermore, aside from the tether, the robot is connected to multiple hoses that replicate the presence of nitrogen and vacuum tubes, which are essential components in real descaling operations. The robot was able to traverse the entire length of the pipe without encountering any issues. The camera output was recorded as the pipe traversal job was unfolding. The robot demonstrated smooth adaptation to varying pipe internal diameters. The articulated mount efficiently rotated during motion without causing any entanglement of the cables and hoses. Moreover, the robot was successfully retrieved without causing damage to any of the cables or hoses, highlighting its precision and safety during the operation.

The current design successfully validates the concept of remote laser descaling, showcasing the robot's capacity to carry the laser and all the required auxiliary hoses through long pipes. This novel method will allow the replacement of existing descaling methods while allowing a safer, more controlled, and long-range scale removal procedure.

## Introduction

Scale deposit poses significant challenges to oil and gas production by obstructing fluid flow to and from wells. This buildup of mineral deposits, along with paraffin, asphaltenes, and debris, gradually narrows the internal diameter of production equipment, thereby reducing flow-rate capacity (Esbai and Palanisamy, 2016; Malik et al., 2017). In severe cases, scaling can completely halt production. For instance, according to Adebayo et. al (Adebayo et. Al, 2020), the scale buildup has led to production stoppage and well abandonment.

To prevent scale buildup, various methods have been developed to descale different types of pipes with varying scale deposits. Numerous studies have focused on mechanical descaling techniques. For instance, Al-Dhafeeri et al. (Al-Dhafeeri et al, 2020), used a torque action debris breaker tool to restore wellbores, demonstrating its effectiveness in removing hard scale and offering cost-effective results compared to other alternatives. Similarly, Adebayo et al. (Adebayo et. al, 2020) conducted underbalanced mechanical descaling operations using coiled tubing, resulting in significant savings, a 30% increase in production, and full restoration of well accessibility.

Chemical descaling is another method developed to remove scaling from pipes. Murtaza et al. (Murtaza et. Al, 2020) proved that a 10.5-pH dissolver efficiently dissolves calcium sulfate scale. He also identified the optimal temperature for scale removal without causing corrosion to metal pipes. Malik et al. (Malik et al., 2018) developed an environmentally friendly descaling solution using an organic acid mixture that dissolves calcareous scale at low pH, reducing pipe corrosion by 39% compared to other chemical descalers.

However, both methods have drawbacks. Mechanical scrubbing can deteriorate the substrate and cause leaks if used frequently (Esbai and Palanisamy, 2016). While chemical pumping is generally more effective, it requires large volumes of treatment chemicals, posing health and safety risks and being impractical for fully plugged deposits.

Recently, Batarseh et al. (Batarseh et al., 2022) developed a high-power laser system for flowline descaling. By optimizing laser parameters like beam size, power, and exposure time, they demonstrated through field tests that the high-power laser could remove different types of scale without damaging the metal substrate. Building on this work, the present study expands the use of the laser system for descaling and enhances its range to reach long distances in the flowline. The laser collimator and auxiliary tubes are mounted on an in-pipe crawler to reach remote parts of the pipe. Additionally, this study examines the design requirements for the robot to traverse the system and provides a detailed explanation of the field test results.

## High power laser for descaling operation

Over the years, high-power lasers have advanced significantly and are now utilized in various fields, including manufacturing, medical, and military applications. Recently, several studies (Batarseh, 2001; Batarseh et al., 2003; Batarseh et al., 2019) have developed a high- power system specifically designed for descaling various pipes used in oil and gas application. Batarseh (Batarseh et al., 2022) conducted a bench test to evaluate the effectiveness of a laser descaling system on flowlines. The system comprises a collimator that focuses a high-power laser beam on the scale, an optical fiber to transmit the optical signal from the laser diode to the collimator, nitrogen hoses to purge the pipe, vacuum hoses, and a process pipe that houses the collimator. This process pipe can move and rotate the collimator to cover the entire perimeter of the pipes. Details of this system are illustrated in Figure 1.

**Figure 1—High-power laser optical components**

The laser system was tested on a 20-foot-long pipe (Batarseh et al., 2022) with varying scale thicknesses, as illustrated in Figure 2-a. The process pipe moves and rotates within the flowline using a linear conveyance system to accurately aim the laser at the scale. Upon completion of the descaling operation, the resulting pipe, shown in Figure 2-b, is scale free, demonstrating the effectiveness of the laser descaling process.


**Figure 2—Pipe joint before (a) and after (b) Laser descaling (Batarseh et al., 2022)**

## Design Overview

To extend the range of the laser descaling system, the process pipe is replaced by a robot able to traverse the pipe while carrying the laser collimator. The designed robot for descaling operations, as shown in Figure 3, utilizes an in-pipe crawler that is designed to be centralized in the pipe. It is connected to a control unit through 100 m spooled tether which defines the reach of the descaling operation. The collimator is attached to an articulated mount that is attached at the end of the in-pipe crawler. The mount maintains the collimator near the edge of the inner diameter of the pipe for smooth descaling operation. The robot's mount is equipped with a motor that facilitates the rotation of the collimator, ensuring coverage of the pipe's inner walls.

Figure 3—Laser descaling robot setup

Additionally, the robot is equipped with various hoses and accessories necessary for laser descaling activities, including nitrogen injection, vacuum cleaning for the removal of scale debris, and water cooling to keep the collimator at an optimal temperature as displayed in Figure 4.

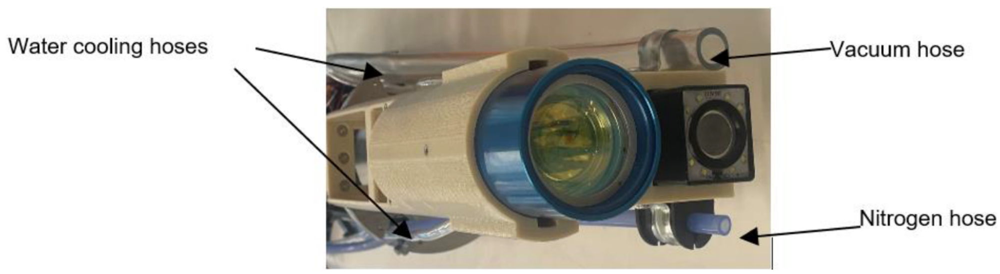


Figure 4—Articulated mount holding different hoses

The control unit includes a graphical interface to manage and oversee the robot's actions. It controls the robot's actuators and speed, and displays sensor outputs such as speed, current, and applied torque on various motors. Additionally, for real-time monitoring of the descaling process, a camera is mounted on the laser mount, allowing close observation of the operation as it unfolds. This setup ensures the efficiency and reliability of the descaling robot. The camera output can also be recorded, enabling follow-up for any descaling job.

## In-Pipe Crawler

The in-pipe crawler, shown in Figure 5, is a robotic system employed to extend the reach of the high-powered laser, enabling long-range descaling up to 100m. its outer diameter (OD) is adjustable, making it adaptable to various pipe sizes, with the ability to expand from 7″ to 10″

The in-pipe crawler incorporates three identical tracks to keep the crawler in a centralized position and prevent the laser collimator from touching the pipe inner diameter. Moreover, the tracks apply enough traction on the pipe to allow the descaling system to overcome minor obstacles and traverse inclined areas while minimizing the risk of surface damage. The tracks are covered with rubber skids allowing them to operate without causing harm to the pipe's surface.

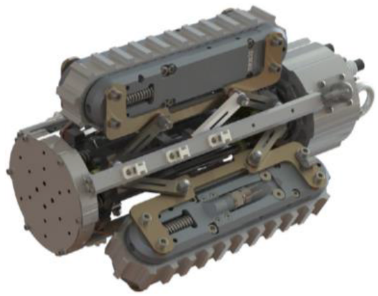

Figure 5—In-Pipe Crawler

The robot is linked to the tether through a waterproof-sealed 18-pin connector at the rear, while the laser is mounted to the front of the robot. In the space between the crawler tracks, hoses and fiber optic cables are fed through to reach the collimator. The robot's structural beams are utilized to secure and prevent the tangling of these hoses and optic cables.

## Articulated mount

The articulated mount is the mechanical component tasked with securely holding the laser, camera, and various hoses, including the nitrogen exhaust pipe and vacuum, as depicted in Figure 6. This mount possesses the capability to adjust the position of the laser collimator, ensuring precise alignment with the scale.

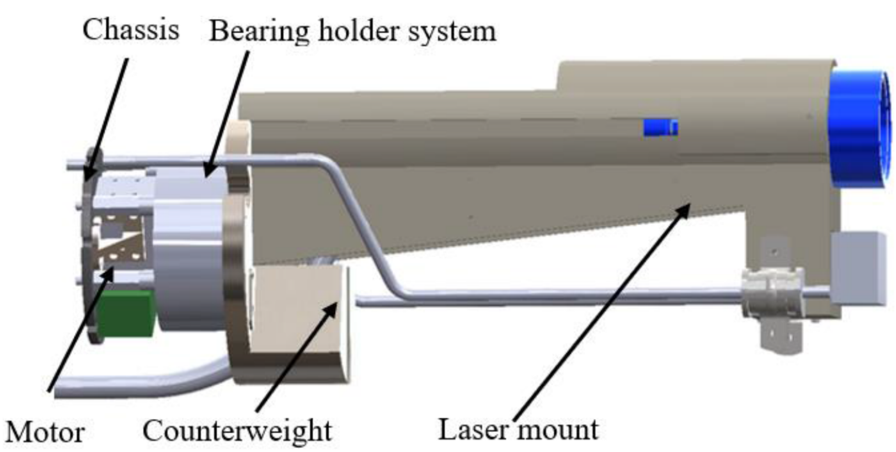


Figure 6—Articulated mount

The mount is made of 5 main parts where each in responsible of a particular function for a precise aiming of the laser collimator

***Chassis.*** This chassis is affixed to the end of the in-pipe crawler. It is engineered to house the motor and the electrical components responsible for rotating the mount.

***Motor.*** The motor selected is a servo motor with high torque capacity and is positioned near its electrical driving circuit. This motor is controlled using the main control unit. The angle and the speed of the motor can be adjusted. Moreover, the motor it can provide information about its status and its current applied torque.

The torque $T$ applied on the motor is determined by the collimator weight of the and the internal diameter (ID) of the pipe as shown in equation 1

$$T = R_{pipe} \times g \times \left(M_{Collimator} + M_{Laser\ mount}\right) \tag{1}$$

Where $R_{pipe}$ is the pipe's inner radius, $g$ is the gravity constant, $M_{Laser\ mount}$ is the mass of the laser mount, and $M_{Collimator}$ is the collimator mass.

***Bearing holder system.*** The bearing holder system is a crucial assembly that connects the motor to the laser mount, eliminates eccentric loads, and facilitates the rotation of the mount via the motor. This assembly consists of two primary components. The first component is a static holder which is affixed to the in-pipe crawler. It contains two bearings minimize the movable plate deflection due to the mount weight and reduce the friction during rotation. The second part is a rotatable plate attached to the motor and remains in constant contact with two bearings to enhance the mobility of the plate.

***Counterweight.*** As depicted in Figure 6, a counterweight is affixed to the rotatable plate in a diametrically opposite position to the collimator mount. The counterweight helps balance the rotating components, reducing vibrations and enhancing safety. It also reduces the total current consumed by the motor. It is carefully designed in terms of weight and placement to offset the torque generated by the combined forces of the laser mount, collimator, and camera.

***Laser mount.*** The laser mount is designed to securely hold the camera, laser collimator, and various hoses in place. The material of the mount must be robust enough to support the weight of the collimator and withstand its operating temperature. At the same time, the mount should be lightweight to ensure that the motors do not require high torque to operate.

The design of this mount is intended to achieve several key objectives like ensuring the laser collimator is positioned as close as possible to the edge of the pipe, providing locations for different hoses to prevent disarray, and accommodating the maximum bend radius of the fiber optic cable.

The distance between the collimator and the camera is carefully configured to ensure that heat generated by the collimator does not adversely affect the camera's operation. To validate this design and assess the camera's exposure to high temperatures during collimator operation, a thermal finite element analysis was conducted on the mount. In this analysis, an extreme case was considered, setting the temperature near the collimator at 200 degrees Celsius.

The results of the analysis, displayed in Figure 7, indicate that, even after 100 seconds, the temperature in the vicinity of the camera remains unchanged, affirming that the camera is not subjected to elevated temperatures.

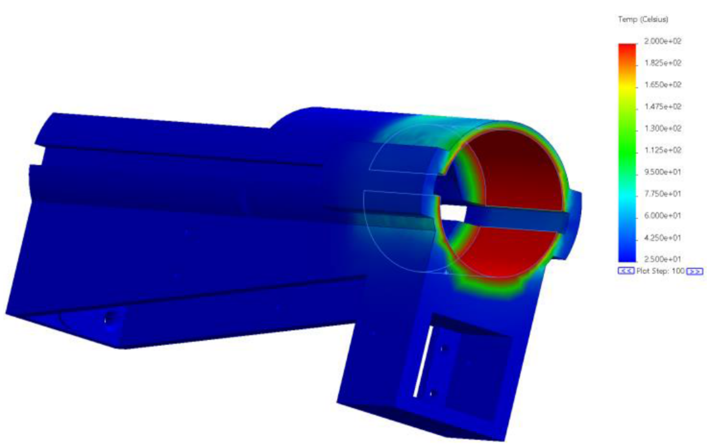


**Figure 7—Heat analyis of the laser mount**

## System limitations

While the design of the descaling robot is intended for a wide range of activities, there are certain limitations that must be considered to ensure an effective operation of descaling.

## Pipe elbow limit angle

When an elbow is present, there is a risk that the laser mount may encounter the inner wall of the pipe as shown in Figure 8. Due to the geometric constraints, there are several positions where the robot can potentially make initial contact with the pipe, and it's important to evaluate the angle limits for these positions. We define α as the elbow angle, which is 0 when the pipe is straight.

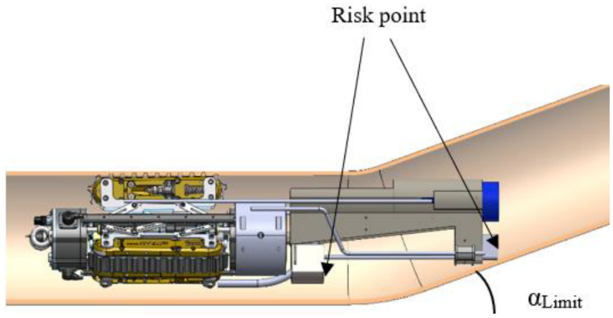


**Figure 8—Descaling robot traversing a pipe elbow**

The analytical equation to determine the maximum allowable elbow angle, $\alpha_{Limit}$, is expressed in Equation 2 as follows:

$$\alpha_{\mathrm{Limit}} = \min\left(\mathrm{atan}\left(\frac{R_p - R_{Co}}{L_{co}}\right), \mathrm{atan}\left(\frac{R_p - R_{Ca}}{L_{Ca}}\right)\right) \tag{2}$$

Here, $R_{Co}$ and $R_{Ca}$ represent the distances between the axis of the robot and the counterweight and the axis of the robot and the camera, respectively. $L_{co}$ and $L_{Ca}$ represent the distances between the end plate and the counterweight end and the distance between the end plate and the camera end. $R_p$ is the inner radius of the pipe.

By utilizing this analysis, a plot of the angle limit in relation to the pipe's inner radius can be generated.

Figure 9 illustrates how the elbow angle varies with the pipe's internal diameter. As anticipated, the angle increases as the pipe ID becomes larger. Notably, there is a distinct shift in the slope when the pipe ID reaches approximately 7.5 inches, which correlates with a shift in the robot's contact point on the pipe. Specifically, for IDs smaller than 7.5 inches, the robot makes contact at the counterweight location, whereas for IDs larger than 7.5 inches, contact occurs at the level of the camera.

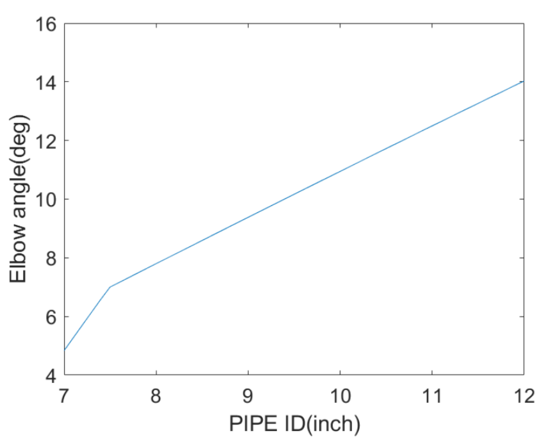


**Figure 9—Variation of elbow angle limit as function of the pipe ID**

## Pipe curvature radius limit

Another feature that can appear on long flowlines is the curvature of the pipes, defined by its radius as depicted in Figure 10. Like elbow deformation, this curvature radius presents a risk of blockage for the descaling robot.

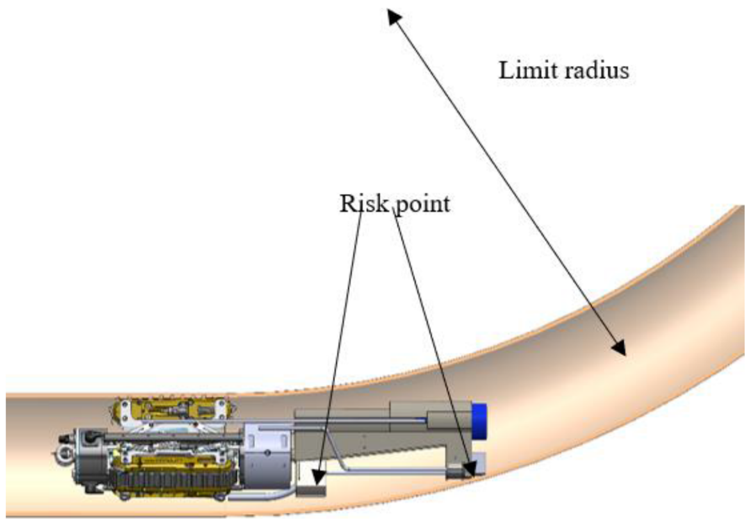


**Figure 10—Descaling robot traversing a curved pipe**

Similar analysis can be performed when the pipe undergoes a curvature and for this analysis the minimum radius of curvature of the pipe can be estimated as shown in equation 3

$$R_{\text{Limit}} = \max\left(\frac{L_{Co}^2}{2(R_p - R_{Co})} + \frac{R_p + R_{Co}}{2}, \frac{L_{Ca}^2}{2(R_p - R_{Ca})} + \frac{R_p + R_{Ca}}{2}\right) \tag{3}$$

Figure 11 shows the evaluation of the maximum radius in relation to the pipe ID. The graph indicates that as the pipe diameter increases, the robot can navigate more acute curvatures without the risk of getting stuck. Additionally, unlike the limit angle curve for the elbow, the curve in Figure 11 is smooth without any slope changes, as the robot's blockage point is only at the camera level.

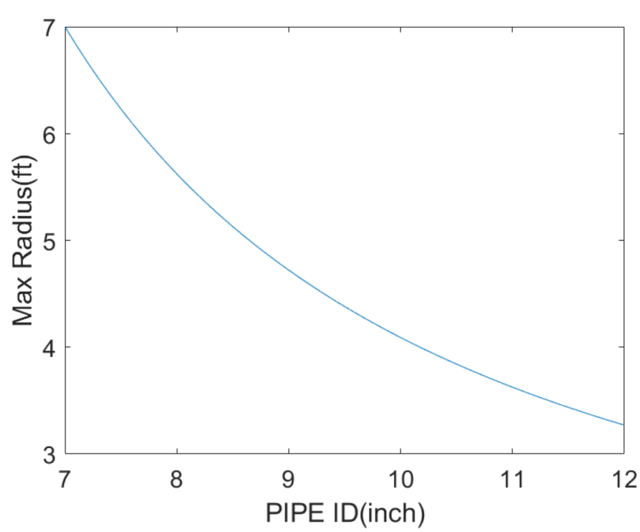


**Figure 11—Variation of pipe curvature radius limit as function of the pipe ID**

### Collimator rotation limit

Because of the presence of the hose and the fiber optic cable, the articulated mount is restricted from making a complete 360-degree rotation. Instead, it is limited to two ranges: 180 degrees from 0 degrees to 180 degrees, and -180 degrees from 0 degrees to -180 degrees, as illustrated in Figure 12

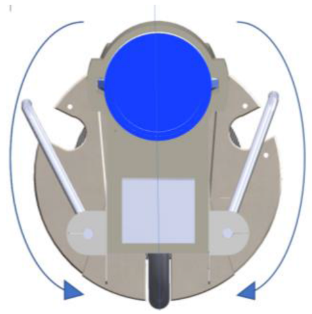

**Figure 12—Rotation limit of the articulated mount**

The rotation pattern of the mount is defined in the code of driving the rotation of the motors. The direction of the rotation for each new input angle is estimated to avoid exceeding the angle limits of the collimator. The speed of rotation is also adjusted to reduce the torque on the collimator and to reduce the induced vibration on the descaling robot.

## Field test

The test setup in the backyard is constructed using plastic pipes that extend over 60m with varying internal diameters as shown in Figure 13. The pipes internal diameter varies form 7.65 inch to 8 inch. These different diameters serve the purpose of demonstrating the descaling robot's ability to navigate through slight changes in internal diameter without compromising stability.

**Figure 13—60m Yard test setup**

Additionally, the pipe setup features a 10m radius curvature to verify the descaling robot's capability to navigate curved paths. Furthermore, aside from the tether, the robot is connected to multiple hoses that replicate the presence of nitrogen and vacuum tubes, which are essential components in real descaling operations.

The test confirmed the robot's ability to traverse the entire length of the pipe without encountering any issues. The robot demonstrated smooth adaptation to varying pipe internal diameters. The articulated mount efficiently rotated during motion without causing any entanglement of the cables and hoses. Moreover, the robot was successfully retrieved without causing damage to any of the cables or hoses, highlighting its precision and safety during the operation.

The descaling robot traversed the 60m pipe within 5 min and reached the end of the designed as shown in Figure 14. The laser descaling was able to traverse the pipe curvature without hitting any of the walls.

**Figure 14—Descaling robot reaching the end of the pipe setup**

## Conclusion

The Descaling robot has been determined to carry the laser collimator, extending its descaling range up to 100 meters within its current scope and potentially up to 1 kilometer in the future. The system demonstrates precision in aiming the laser collimator while traversing the pipe. Through conducted tests, the robot exhibited the ability to navigate long pipes, various elbows, radius bends, and changes in pipe thickness. The current design successfully validates the concept of remote laser descaling, showcasing the robot's capacity to carry the laser and all the required auxiliary hoses through long pipes.

The current design can be extended to support both flowline and wireline descaling. Additionally, incorporating more sensors into the system can further optimize the descaling process.

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